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Antenna modeling and optimization based on 1D GP-HetCNN surrogates and intelligent methods

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Abstract

This paper proposes a lightweight antenna surrogate model based on a one-dimensional grouped-and-pointwise heterogeneous convolutional neural network (1D GP-HetCNN) for the prediction of the scattering parameter (S_{11}) curve of antennas. Using antenna physical parameters as the input, the surrogate model can predict the S_{11} curve efficiently and accurately. Then, in order to study complicated frequency features of curves, the 1D GP-HetCNN model is further combined with a multi-task learning method, constructing 1D GP-HetCNN-MTL. Based on the two trained surrogate models, two optimization methods employing the whale optimization algorithm and the coati optimization algorithm are developed respectively. On the test set, the 1D GP-HetCNN model achieves the root mean square error (RMSE) of 0.0305 and the mean absolute error (MAE) of 0.0189; the 1D GP-HetCNN-MTL model achieves the RMSE of 0.0426 and the MAE of 0.0282. Both surrogate models perform better than the two baseline models. Moreover, results from the two optimization methods show that they can identify improved physical parameters with a wider operating bandwidth from 19.3 GHz to averagely 20.0 GHz more efficiently than conventional electromagnetic (EM) simulation, which increases the fractional bandwidth from 155.02% to averagely higher than 156%. These findings indicate that surrogate models combined with intelligent optimization algorithms can accelerate antenna modeling and further improve the antenna's EM performance. Overall, the proposed approach facilitates more efficient antenna design and contributes to the development of intelligent antenna engineering.

Keywords Convolutional neural network, Antenna modeling, Surrogate model, Multi-task learning, Intelligent optimization algorithm

1 Article highlights

Two compact neural surrogate models accurately predict an antenna's input-reflection response across the full design band. By replacing time-consuming full-wave simulations with fast surrogate evaluations (seconds per candidate), the models allow designers to explore orders of more geometric variants during a single design cycle. This rapid



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feedback shortens iteration time, lowers computational cost, and makes broad design exploration and sensitivity studies feasible within routine engineering schedules.

Coupling the surrogates with two complementary intelligent optimizers produces automated searches that find improved antenna geometric parameters with wider usable bandwidth. The surrogates steer the optimizers so they can evaluate many candidate solutions cheaply; the optimizers then focus on promising regions of the design space. While heuristic searches do not guarantee a global optimum, this surrogate-assisted pipeline substantially raises the chance of locating high-quality, practically usable designs with far fewer expensive simulations and less manual trial-and-error.

The surrogate-assisted workflow balances speed and reliability by using the fast models for large-scale search and reserving costly high-fidelity simulations for final validation. This staged approach preserves the trustworthiness of physics-based verification while dramatically cutting the number of full-wave runs needed. The outcome is reduced development time, lower compute expense, and less human effort—facilitating quicker translation from simulation to prototype and making the method suitable for both research and industrial antenna design practice.

2 Introduction

The extensive utilization of antennas in the 5 G, 6 G, millimeter wave, and terahertz (THz) communication domains has led to the emergence of increasingly sophisticated and high-performance antenna design requirements [1, 2], which has posed significant challenges, including a substantial theoretical foundation in electromagnetism, proficiency in simulation software and high time cost to conventional design methods [3], impeding the rapid design and iteration process of antennas, and necessitating the development of advanced methodologies to address these issues. Therefore, there is an urgent need for more automated, intelligent, and cost-effective design methods.

In recent years, deep learning (DL) technology has made significant strides in numerous domains, including medicine, autonomous driving, and mechanical manufacturing [4–6], and has also been gradually applied in electromagnetism, thereby stimulating research and development in the field of smart antenna design.

The integration of DL has enabled the establishment of intricate nonlinear mapping relationships between antenna physical parameters and electromagnetic (EM) response, offering a potential alternative to conventional simulation methods.

Deep neural networks (DNNs) of DL have received particular attention in tasks such as antenna design [7–9], antenna optimization [10–12], and antenna fault detection [13–15], because their deeper architectures enables the extraction of more complex feature information, facilitating a more precise mapping between the physical parameters of the antenna and the EM response.

DNNs have demonstrated significant potential in the realm of antenna design. However, the construction of high-precision DNN models frequently necessitates the accumulation of abundant training data and the iterative refinement of model hyperparameters to ensure its generalizability.

Nevertheless, the process of obtaining sufficient high-quality training data has become a major challenge in applying DNN models to antenna design because of the high time and computational costs of high-precision EM simulation [16].

Therefore, the integration of intelligent optimization algorithms with DNNs has emerged as a novel solution for antenna design. These algorithms possess a robust global search capability and a swift convergence rate, enabling autonomous adjustment of the DNN model's hyperparameters, enhancement of its automation level, and reduction of manual intervention.

Furthermore, intelligent optimization algorithms can be utilized in conjunction with a trained antenna surrogate model to automatically search for antenna physical parameters that match the EM targets, thereby markedly reducing the necessity for repeated experiments.

Moreover, the hybrid intelligent optimization method has been shown to effectively mitigate the probability of the model converging on a local optimal solution, thus enhancing the EM response while adhering to the stipulated design requirements [16–18].

To achieve fast and accurate antenna scattering parameter (S_{11}) curve prediction and optimize antenna performance, this paper proposes a lightweight DL surrogate model based on 1D GP-HetCNN. Taking the antenna's physical parameters as the input, the model can efficiently predict the S_{11} curve. Then, the model is further combined with a multi-task learning method which constructs 1D GP-HetCNN-MTL, to study complicated frequency features of S_{11} curves.

The better physical parameters are subsequently searched through a respective combination of two surrogate models, the whale optimization algorithm (WOA) [19], and the coati optimization algorithm (COA) [20], with the objective of expanding the antenna's operating bandwidth.

A miniaturized UWB antenna [21] is selected as the application case during the whole process.

The remainder of this paper is organized as follows. In Sect. 2, the proposed surrogate models and intelligent optimization methods are introduced in detail, including the construction of the surrogate models and the application of the optimization methods. Section 3 presents and analyzes the application process and results. The final section provides the conclusions and future work of our comprehensive study.

3 Proposed surrogate model and intelligent optimization methods

3.1 Construction of the surrogate model

Since LeNet and MNIST recognition [22], convolutional neural networks (CNNs) have evolved into diverse variants, with the innovation of these network structures promoting the wide application of CNNs in various fields. The strong learning ability of CNNs is also evident in antenna modeling, optimization, and fault detection [9, 12, 13].

However, as the depth of the network increases, the homogeneous design of the traditional CNN restricts the network's capacity to process diverse local features at multiple scales, leading to excessive computation, augmented computing memory consumption.

Consequently, recent innovations in CNNs, including heterogeneous convolution (HetConv) [23] and dynamic convolution [24], have emerged to increase the ability of the network to capture features across various scales, thereby enhancing the model's performance. These enhancements enable the model to extract key features with greater precision and to fulfill a range of complicated tasks while preserving efficiency.

This paper proposes a lightweight surrogate model, 1D GP-HetCNN, based on grouped convolution and cross-channel fusion, and the architecture is illustrated in Fig. 1a.

To enhance the feature extraction capability while reducing the computational complexity of conventional CNN layers, a 1D grouped-and-pointwise HetConv (GP-HetConv) module is proposed. Each 1D GP-HetConv module contains two parallel branches: groupwise convolution (GWC) and pointwise convolution (PWC).

The GWC implements a channel grouping strategy. The first GWC (Fig. 1b) uses a group number $P = 1$, because the number of the input channel is 1; whereas subsequent GWCs (Fig. 1c) adopt $P = 4$. Each group employs 5×1 kernels to extract local features with extended receptive fields.

Meanwhile, the PWC of each 1D GP-HetConv module utilizes 1×1 kernel to fuse cross-channel information.

Finally, the outputs of both branches are element-wise summed as Eq. (1). This operation combines local detailed features with cross-channel correlations, mitigating the limitation of single-kernel designs.

$$\text{Output} = \text{GWC}(x) + \text{PWC}(x). \quad (1)$$

Unlike the original HetConv which requires heterogeneous kernels across groups, the GWC of 1D GP-HetConv adopts uniform 5×1 kernels for all groups, simplifying implementation. Moreover, instead of concatenating multi-scale outputs from heterogeneous kernels, 1D GP-HetConv enhances feature representation through complementary interactions between GWC and PWC.

Compared with that of conventional CNN layers, the computational efficiency of the 1D GP-HetConv modules is significantly greater.

Firstly, the parameter count N of a conventional convolution block is defined as Eq. (2):

$$N_{\text{conv}} = \text{in-ch} \times \text{out-ch} \times k, \quad (2)$$

where in-ch represents the number of input channels, out-ch represents the number of output channels, and k represents the kernel size.

The parameter count of a GWC with g groups is defined as Eq. (3):

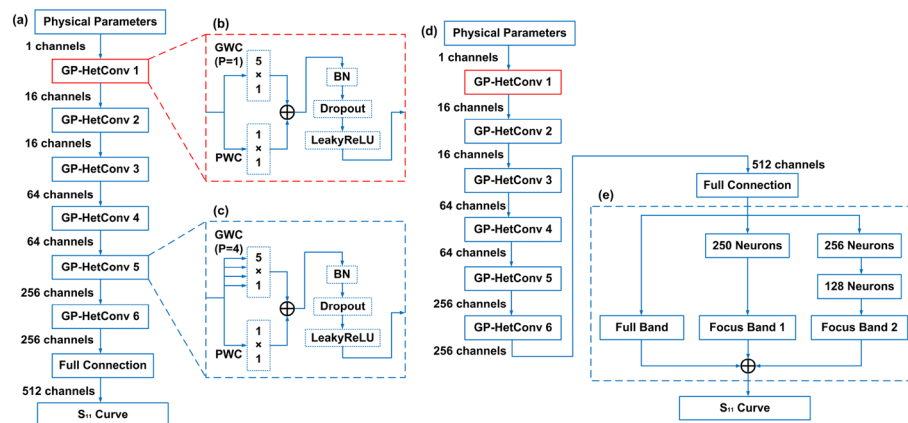


Fig. 1 Architecture of proposed models: (a) architecture of 1D GP-HetCNN; (b) structure of 1D GP-NetConv 1 module; (c) structure of other 1D GP-NetConv modules; (d) architecture of 1D GP-HetCNN-MTL; (e) structure of the MTL method

$$N_{gwc} = \frac{in-ch \times out-ch \times k}{g}, \quad (3)$$

and the parameter count of a GWC is defined as Eq. (4):

$$N_{pwc} = in-ch \times out-ch. \quad (4)$$

Hence, the parameter count of a 1D GP-HetConv block is defined as Eq. (5):

$$N_{GP} = N_{gwc} + N_{pwc} = in-ch \times out-ch \left(\frac{k}{g} + 1 \right). \quad (5)$$

Taking the second convolution block ($in-ch=16$, $out-ch=16$) as an example, N_{conv} is 1280, whereas N_{GP} is 576, which achieves 55% parameter reduction.

Secondly, the floating point operation (*FLOP*) between the conventional CNN layers and the 1D GP-HetConv modules is compared. The *FLOP* is defined in equation set (6):

$$\begin{aligned} FLOP_{conv} &= in-ch \times out-ch \times k \times L \times 2, \\ FLOP_{GP} &= \left(\frac{in-ch \times out-ch \times k}{g} + in-ch \times out-ch \right) \times L \times 2, \end{aligned} \quad (6)$$

where $in-ch$ represents the number of input channels, $out-ch$ represents the number of output channels, k represents the kernel size, L represents the length of the input sequence, and $\times 2$ accounts for multiply-accumulate operations.

For the second convolution block, the $FLOP_{conv}$ is $2560L$, whereas the $FLOP_{GP}$ is $1152L$ which achieves a 55% reduction.

These two findings confirm that the 1D GP-HetConv modules maintain feature extraction capacity while drastically reducing computational costs.

To further enhance the performance of the surrogate model, a series of optimization strategies have been employed, including the implementation of LeakyReLU as the activation function, which retains a gradient when the input is less than zero, thereby alleviating the potential neuron death problem of ReLU and ensuring more stable gradient flow.

Additionally, batch normalization (BN) has been utilized to accelerate training, reducing internal covariate shifts and improving the generalization ability of the model.

Moreover, the standard pooling layer is replaced by global adaptive average pooling, a method capable of automatically adjusting the size of the pooling window. This modification results in a fixed final output feature map size, a reduction in model parameters, and the ability to learn information at varying scales.

Dropout (dropout-rate = 0.2) is incorporated into the fully connected layer with the objectives of reducing overfitting and enhancing the model's robustness when confronted with complex tasks.

Multi-task learning (MTL) [25], a paradigm that jointly learns multiple related tasks by sharing intermediate representations, has achieved significant success in various domains of DL, with recent applications extending to antenna design and optimization [26, 27].

Different frequency ranges of S_{11} curves often exhibit distinct local behaviours (e.g., resonances and troughs). To extract more features of curves, the full-band S_{11} and two

engineering-relevant focus bands (focus1 and focus2) are treated as three parallel tasks within the surrogate model, as illustrated in Fig. 1d.

Specifically, the model employs a shared 1D GP-HetConv-based backbone with an initial full connection layer and three task-specific output heads (Fig. 1e) that predict the full-band and the two sub-band S_{11} vectors. The design of the three heads is based on the length of three task vectors. In implementation, the output dimensions are denoted D_{full} , D_{f1} , D_{f2} for the full band, focus1 and focus2, respectively; the mapping between full-band indices and sub-band slices is defined consistently in the model.

Training adopts the standard hard-parameter-sharing MTL scheme. The shared backbone learns a common representation from the physical input, while each task-specific head refines that representation for its assigned frequency range. Denoting the model outputs for a given batch as $\hat{y}_{full} \in \mathbb{R}^{D_{full}}$, $\hat{y}_{f1} \in \mathbb{R}^{D_{f1}}$ and $\hat{y}_{f2} \in \mathbb{R}^{D_{f2}}$ with corresponding ground-truth vectors y_{full} , y_{f1} , y_{f2} . For each task, the SmoothL1Loss (Huber-like) is adopted, and the per-task losses are defined as equation set (7).

$$\begin{aligned} L_{full} &= \text{SmoothL1}(\hat{y}_{full}, y_{full}), \\ L_{f1} &= \text{SmoothL1}(\hat{y}_{f1}, y_{f1}), \\ L_{f2} &= \text{SmoothL1}(\hat{y}_{f2}, y_{f2}). \end{aligned} \quad (7)$$

To allow explicit emphasis on engineering-critical frequency regions, a weighted joint loss is used as (8):

$$L_{total} = \alpha_{full} L_{full} + \alpha_{f1} L_{f1} + \alpha_{f2} L_{f2}, \quad (8)$$

where α_* are non-negative task weights chosen according to the design priorities.

Importantly, treating full-band and sub-band predictions as separate tasks does not by itself sever inter-band correlations. Because the backbone is shared and is updated by gradients derived from all tasks, it learns features that reflect cross-frequency dependencies, while the task-specific heads perform local refinement on that shared representation.

In summary, the MTL design adopted here combines a 1D GP-HetConv-based shared backbone with task-weighted joint training to preserve cross-band information while trying to improve prediction fidelity in engineering-relevant frequency regions.

3.2 Intelligent optimization methods

3.2.1 Whale optimization algorithm

The WOA [19] is a population-based, bio-inspired metaheuristic proposed by Mirjalili in 2016. It models certain foraging behaviors of humpback whales (notably the bubble-net feeding strategy) and has been applied in many engineering optimization tasks, including antenna design [28, 29].

WOA represents each search agent (whale) as a candidate solution $x \in \mathbb{R}^d$ in the design space. The algorithm alternates between exploitation and exploration via two principal mechanisms:

-Encircling / shrinking behaviour (exploitation). The current best solution x^* is assumed to be the prey. Defining equation set (9),

$$\begin{aligned} A &= 2ar_1 - a, \\ C &= 2r_2, \end{aligned} \quad (9)$$

where a decreases linearly from 2 to 0 over iterations, and $r_1, r_2 \in [0, 1]^d$ are random vectors.

Then, defining equation set (10),

$$\begin{aligned} D &= |C \odot x^* - x|, \\ x(t+1) &= x^* - A \odot D, \end{aligned} \quad (10)$$

which is used when $|A| < 1$ to encourage exploitation around the best-so-far.

-Spiral update (exploitation). With probability p , a spiral model approximates the bubble-net motion as equation (11):

$$x(t+1) = D' \exp(b\ell) \cos(2\pi\ell) + x^*, \quad (11)$$

where $D' = |x^* - x|$, b is a constant controlling the spiral shape, and $\ell \sim \mathcal{U}(-1, 1)$.

-Exploration (random search). When $|A| \geq 1$, an agent is updated relative to a randomly selected agent x_{rand} as equation set (12):

$$\begin{aligned} D_{rand} &= |C \odot x_{rand} - x|, \\ x(t+1) &= x_{rand} - A \odot D_{rand}. \end{aligned} \quad (12)$$

A typical implementation uses $p = 0.5$, $b = 1$, and a linear schedule for a . Note that WOA is stochastic and heuristic: while it often exhibits strong empirical global-search capabilities and relatively fast convergence on many engineering problems, it does not provide a mathematical guarantee of finding the global optimum.

In the surrogate-assisted pipeline, WOA operates in the normalized design space (min-max scaling consistent with surrogate training). At each iteration, the population of N_{pop} agents is evaluated in batch by the surrogate (single batched forward pass) to compute scalar fitness values; updates are then performed according to the WOA rules, and candidate vectors are clipped or repaired to satisfy bounds. The main computational cost per generation is therefore N_{pop} surrogate evaluations (plus negligible update overhead).

3.2.2 Coati optimization algorithm

The COA [20] is a recent population-based metaheuristic inspired by the social and evasive behaviours of coatis. COA alternates between global exploration (a hunting/foraging phase that moves individuals into different regions of the search space) and local exploitation (an evasion/retreat phase that concentrates search near promising solutions). Like other swarm-based metaheuristics, COA is derivative-free and stochastic; it is therefore suitable for black-box optimization problems such as surrogate-assisted antenna design.

A practical COA implementation maintains a population of N_{pop} agents $\{x_i(t) \in [0, 1]^d\}_{i=1}^{N_{pop}}$ at generation t , where d denotes the design dimensionality. The algorithm iteratively updates agent positions using a combination of attack-like moves (moving toward a leader or best-so-far solution $x^*(t)$) and spiral/retreat moves (local refinement around leaders). In the implementation used here, a linear schedule parameter $a(t)$ that decays from 2 to 0 over T iterations is employed as Eq. (13):

$$a(t) = 2\left(1 - \frac{t}{T-1}\right), \quad t = 0, \dots, T-1. \quad (13)$$

For each agent, the random vectors are drawn as $r_1, r_2 \sim \mathcal{U}(0, 1)^d$ and form the auxiliary vectors A and C as Eq. (14):

$$A = 2a(t)r_1 - a(t), \quad C = 2r_2, \quad (14)$$

where $A, C \in \mathbb{R}^d$. Denoting the spiral probability by $p_{spiral} \in [0, 1]$ and the spiral shape constant by $b > 0$.

With probability p_{spiral} , the agent performs a spiral/local refinement around the leader as Eq. (15):

$$\begin{aligned} \ell &\sim \mathcal{U}(-1, 1)^d, \\ D' &= \|x^*(t) - x_i(t)\|_2, \\ x_i(t+1) &= x^*(t) + D' \odot \exp(b\ell) \odot \cos(2\pi\ell), \end{aligned} \quad (15)$$

where D' is a scalar and the scalar-vector multiplication $D' \odot (\odot)$ denotes broadcasting of D' across vector components.

Otherwise (with probability $1 - p_{spiral}$), the agent executes a leader-following or exploration update. Using an elementwise-based exploitation criterion implemented via the infinity norm:

$$\text{if } \|A\|_\infty < 1 \quad (\text{i.e. } \max_j |A_j| < 1),$$

then perform the encircling (leader-following) update as equation set (16):

$$\begin{aligned} D &= |C \odot x^*(t) - x_i(t)|, \\ x_i(t+1) &= x^*(t) - A \odot D, \end{aligned} \quad (16)$$

else perform exploration relative to a randomly selected peer $x_{rand}(t)$ as equation set (17):

$$\begin{aligned} D_{rand} &= |C \odot x_{rand}(t) - x_i(t)|, \\ x_i(t+1) &= x_{rand}(t) - A \odot D_{rand}. \end{aligned} \quad (17)$$

After each update, projecting to the feasible normalized box via elementwise clipping as Eq. (18):

$$x_i(t+1) \leftarrow \text{clip}(x_i(t+1), 0, 1), \quad (18)$$

i.e. $(\text{clip}(v, 0, 1))_j = \min(\max(v_j, 0), 1)$.

COA is stochastic and heuristic: while it can be effective in practice, it does not provide mathematical guarantees of global optimality. In the surrogate-assisted pipeline, COA operates on normalized design vectors. At each generation, we form the population matrix $X(t) \in \mathbb{R}^{N_{pop} \times d}$ whose i -th row is $x_i(t)^\top$. The surrogate $\mathcal{S}$ is evaluated in batch to produce population outputs as Eq. (19):

$$\{\hat{y}_i\}_{i=1}^{N_{pop}} = \mathcal{S}(X(t)), \quad (19)$$

where $\hat{y}_i \in \mathbb{R}^{N_{out}}$ denotes the surrogate-predicted S_{11} vector for agent i . A scalar fitness is computed for each agent by an engineering objective $f: \mathbb{R}^{N_{out}} \rightarrow \mathbb{R}$ as Eq. (20):

$$s_i = f(\hat{y}_i), \quad i = 1, \dots, N_{pop}. \quad (20)$$

Positions are updated according to Eqs. (15)–(17), clipped by Eq. (18), and the best-so-far is recorded as Eq. (21):

$$x^*(t+1) = \arg \max_i s_i. \quad (21)$$

Notation: the surrogate-assisted optimization experiments reported in this paper use two population-based metaheuristics: the WOA [19] and the COA [20]. The choice was made according to explicit pragmatic and scientific criteria rather than an assertion of universal superiority. We summarise these criteria below.

- 1) Complementary exploration–exploitation dynamics: WOA implements a contraction/spiral exploitation mechanism centred on the current best solution, while COA employs cooperative hunting followed by evasion/retreat phases. Selecting one algorithm from each family exposes the surrogate-optimizer pipeline to different search dynamics (spiral refinement vs. cooperative/escape behaviour), helping to assess robustness across algorithm classes.
- 2) Parameter parsimony and practical tuning: both WOA and COA are relatively parameter-light (few interpretable hyperparameters, such as the linear a -schedule and spiral constant b). When surrogate evaluations dominate computational cost, reduced hyperparameter tuning burden is an important practical advantage.
- 3) Batch-friendly, low per-iteration overhead: The core update formulas used by WOA and COA are algebraically simple and naturally vectorisable, which allows evaluating an entire population with a single batched forward pass through the neural surrogate. This vectorisation reduces wall-clock overhead per generation and was a key practical consideration given the surrogate-evaluation budget used in our comparisons.
- 4) Representative benchmarking: WOA has an established record in continuous EM and antenna optimization problems and serves as a mature representative of spiral/contraction heuristics. COA is a recent cooperative heuristic whose group-based mechanics provide a complementary search paradigm. Including both methods demonstrates the pipeline’s applicability to distinct optimizer families.

We emphasise that other well-known algorithms are valid alternatives and can be integrated in the same surrogate-assisted workflow; the Bayesian optimization (BO) algorithm, the particle swarm optimization (PSO) algorithm and the gray wolf optimization (GWO) algorithm are included among our baselines.

3.2.3 Objective function

For the surrogate based on 1D GP-HetCNN, the scalar fitness is computed directly from the model’s full-band S_{11} prediction. Let $\hat{S}_{11}^{full} \in \mathbb{R}^{D_{full}}$ denote the denormalized predicted full-band vector sampled at the discrete frequency set $\mathcal{F} = \{f_1, \dots, f_{D_{full}}\}$. The base margin is defined as Eq. (22):

$$Base\ Margin_{full} = 10 - \max_{f \in \mathcal{F}} \hat{S}_{11}^{full}(f). \quad (22)$$

Using a threshold $T = -10$ dB, the (primary) contiguous index span $[i_{first}, i_{last}]$ where $\hat{S}_{11}^{full}(f) \leq T$ is located, and a within-span penalty is computed by summing positive exceedances above the threshold as Eq. (23):

$$Penalty_{full} = \sum_{i=i_{first}}^{i_{last}} \max(0, \hat{S}_{11}^{full}(f_i) - T). \quad (23)$$

The scalar objective, $Final\ Margin_{full}$, returned to the optimizer is then set as Eq. (24):

$$Final\ Margin_{full} = Base\ Margin_{full} - \lambda \cdot Penalty_{full}, \quad (24)$$

where $\lambda \geq 0$ is the user-selected penalty weight. The optimizer seeks to maximize $Final\ Margin_{full}$.

The pseudocode for calculating the margin and penalty term based on 1D GP-HetCNN is provided as Algorithm 1.

Algorithm 1 Physical Parameter Searching Based on single-head 1D GP-HetCNN

```

1: Input: Normalized Physical Parameters: {param0, ..., param7}
2: Output: Final Margin
3: geom  $\leftarrow$  [parameters[parami] for  $i \in 0 \dots 7$ ]
4: x  $\leftarrow$  ToTensor(geom).reshape(1, 1, -1).to(device)
5: full  $\leftarrow$  GP_HetCNN(x) ▷ full-band prediction (de-normalized)
6: region  $\leftarrow$  full
7: base_margin  $\leftarrow$  10 - max(region)
8: function COMPUTE_PENALTY(region, threshold = -10)
9:   indices  $\leftarrow$  where(region < threshold)[0]
10:  if indices.size > 0 then
11:    first  $\leftarrow$  indices[0]
12:    last  $\leftarrow$  indices[end]
13:    subregion  $\leftarrow$  region[first : last + 1]
14:    violations  $\leftarrow$  max(0, subregion - threshold)
15:    return sum(violations)
16:  else
17:    return 0
18:  end if
19: end function
20: return base_margin -  $\lambda \times$  compute_penalty(region)

```

The 1D GP-HetCNN-MTL produces three de-normalized outputs: a full-band vector $\hat{S}_{11}^{full}$ and two focus-band vectors $\hat{S}_{11}^{f1} \in \mathbb{R}^{D_{f1}}$, $\hat{S}_{11}^{f2} \in \mathbb{R}^{D_{f2}}$. For engineering prioritization, per-focus-band margins are computed and combined as the base margin. Specifically, the per-band base margins are defined as equation set (25):

$$\begin{aligned} Base\ Margin_{f1} &= 10 - \max_{f \in \mathcal{F}_{f1}} \hat{S}_{11}^{f1}(f), \\ Base\ Margin_{f2} &= 10 - \max_{f \in \mathcal{F}_{f2}} \hat{S}_{11}^{f2}(f), \end{aligned} \quad (25)$$

and the combined base margin defined as Eq. (26) used in the optimizer is the arithmetic mean of the two focus margins:

$$\begin{aligned} Base\ Margin_{MTL} &= \frac{Base\ Margin_{f1} + Base\ Margin_{f2}}{2} \\ &= \frac{20 - \max(\hat{S}_{11}^{f1}) - \max(\hat{S}_{11}^{f2})}{2}. \end{aligned} \quad (26)$$

Penalties for each focus band are computed in the same manner as in the single-head case (sum of positive exceedances inside the primary below-threshold contiguous span). Denoting these penalties by $Penalty_{f1}$ and $Penalty_{f2}$, the final scalar objective is defined as Eq. (27):

$$Final\ Margin_{MTL} = Base\ Margin_{MTL} - \lambda \cdot (Penalty_{f1} + Penalty_{f2}). \quad (27)$$

The pseudocode for calculating the margin and penalty term based on 1D GP-HetCNN-MTL is provided as Algorithm 2.

Algorithm 2 Physical Parameter Searching Based on 1D GP-NetCNN-MTL

```

1: Input: Normalized Physical Parameters: {param0, ..., param7}
2: Output: Final Margin
3: geom  $\leftarrow$  [parameters[parami] for  $i \in 0 \dots 7$ ]
4:  $\mathbf{x} \leftarrow \text{ToTensor}(\text{geom}).\text{reshape}(1, 1, -1).\text{to}(\text{device})$ 
5: ( $\cdot$ , focus1, focus2)  $\leftarrow \text{GP\_HetCNN\_MTL}(\mathbf{x})$ 
6: region1  $\leftarrow \text{focus}_1 \times (\text{max}_1 - \text{min}_1) + \text{min}_1$ 
7: region2  $\leftarrow \text{focus}_2 \times (\text{max}_2 - \text{min}_2) + \text{min}_2$ 
8: base_margin  $\leftarrow \frac{20 - \max(\text{region}_1) - \max(\text{region}_2)}{2}$ 
9: function COMPUTE_PENALTY(region, threshold = -10)
10:   indices  $\leftarrow \text{where}(\text{region} < \text{threshold})[0]$ 
11:   if indices.size > 0 then
12:     first  $\leftarrow \text{indices}[0]$ 
13:     last  $\leftarrow \text{indices}[\text{end}]$ 
14:     subregion  $\leftarrow \text{region}[\text{first} : \text{last} + 1]$ 
15:     violations  $\leftarrow \max(0, \text{subregion} - \text{threshold})$ 
16:     return sum(violations)
17:   else
18:     return 0
19:   end if
20: end function
21: return base_margin -  $\lambda \times (\text{compute\_penalty}(\text{region}_1) + \text{compute\_penalty}(\text{region}_2))$ 

```

4 Application case of modeling and optimization

4.1 Antenna case

The UWB stepped microstrip monopole antenna [21] employed as the case is depicted in Fig. 2a,b. The antenna's dimensions are 13 mm $\times$ 22 mm $\times$ 0.8mm, and it is fed by

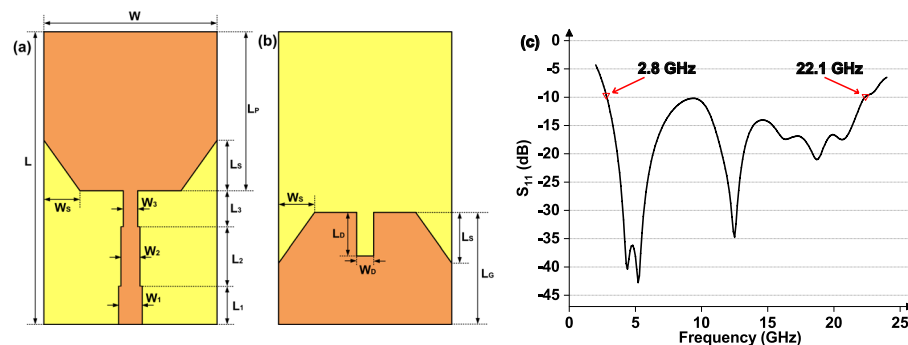


Fig. 2 Antenna structure and S_{11} curve: (a) the top structure of the antenna; (b) the bottom structure of the antenna; (c) S_{11} simulated curve from HFSS

Table 1 Specific physical parameters of the antenna (mm)

Parameter	Value	Parameter	Value	Parameter	Value
L	22.0	W	13.0	H	0.8
W_1	1.5	W_2	1.3	W_3	1.0
L_1	3.0	L_2	5.0	L_3	2.0
L_P	12.0	L_S	4.216	W_S	3.5
L_G	8.7	L_D	3.6	W_D	1.295

Table 2 Sampling details of the physical parameters

Parameter	Scale (mm)	Step (mm)	Number
L_D	3.6–3.7	0.1	2
L_G	8.6–8.7	0.1	2
L_S	4.108–4.216	0.108	2
W_1	1.4–1.6	0.1	2
W_2	1.2–1.3	0.05	3
W_3	0.9–1.0	0.1	2
W_D	1.27–1.295	0.025	2
W_S	3.4–3.5	0.1	2

a stepped microstrip line. The radiation patch is a rectangular structure with a triangular missing slot, and the defected ground structure (DGS) is employed to optimize the performance of the antenna. The comprehensive physical parameters are provided in Table 1.

By utilizing the High Frequency Structure Simulator (HFSS) for the full-parameter scanning simulation, the S_{11} curve of the antenna is depicted in Fig. 2c. The simulation outcomes demonstrate that the proposed antenna complies with the operational requirements within the frequency band spanning from 2.8 to 22.1 GHz.

The eight key physical parameters are denoted as $G = [L_D, L_G, L_S, W_1, W_2, W_3, W_D, W_S]$, and the corresponding value range and sampling step are set for each parameter, as shown in Table 2.

The construction of the dataset is achieved through the full-parameter scanning simulation of HFSS. The process yields a total of 576 sets of data, each comprising eight physical parameters and their respective S_{11} curve sampling sequences.

The frequency range of the S_{11} curve is set to be 2 GHz to 24 GHz, and the sampling step size is set to 0.1 GHz, resulting in each S_{11} curve sampling sequence containing 221 sampling values.

Partial S_{11} curves in the dataset are shown in Fig. 3. The distribution of all the S_{11} curves simulated by the HFSS reveals a relatively consistent fluctuation trend between 2 GHz and 17 GHz, whereas after 17 GHz, the fluctuation trend of the S_{11} curves differs significantly, indicating that the high-frequency band region, i.e., above 17 GHz, is the domain in which the surrogate model needs to focus on learning.

In light of these findings, 2.6 GHz to 17 GHz and 17.1 GHz to 22.1 GHz are selected as the two focus learning bands to train 1D GP-HetCNN-MTL. The full frequency band tasks, ranging from 2 to 24 GHz, are also incorporated into the framework to ensure the correlation of S_{11} values at different frequency points. The weights of the loss function are set to $\alpha_{full} = 0.1$, $\alpha_{focus1} = 0.4$ and $\alpha_{focus2} = 0.5$ for the three tasks.

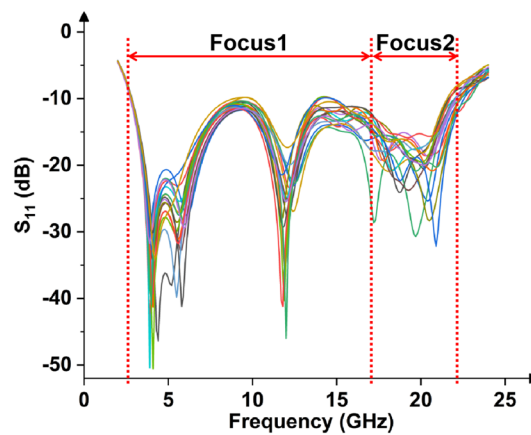


Fig. 3 Partial S_{11} curves in dataset

4.2 Training strategies and experimental process

To enhance the predictive capability of the surrogate models for the S_{11} curve, it is imperative to undertake extensive training to minimize the error between the predicted curve and the actual curve. In this paper, the maximum number of training epochs is set to 1000 to ensure that the model can thoroughly learn complex patterns in the data. Several strategies can be employed to increase prediction accuracy during model training.

Firstly, the input physical parameters and the output S_{11} sampling sequences are normalized to the interval $[0, 1]$ by Min-Max normalization to reduce the influence of different dimensions on the training process and improve the numerical stability of the model.

Then, the dataset is divided into a training set and a test set at a ratio of 8 : 1 to ensure that the model can learn effectively during the training process, and the generalization ability is evaluated on the test set.

In the training process, the Adam optimizer is used to update the gradient, taking into account the adaptive learning rate (lr). This approach is employed to enhance the convergence speed and prevent the issues of gradient disappearance or gradient explosion.

The initial lr is set to $1e-3$, and the ReduceLROnPlateau learning rate scheduling strategy is introduced to dynamically decrease the lr during the training process. This strategy aims to improve the convergence stability and generalization ability of the model. The lr decay factor is set to 0.1, and the patience value is set to 50 epochs. That is, when the test set loss does not decrease significantly within 50 epochs, the lr will be attenuated according to the set factor.

Additionally, SmoothL1Loss is employed as the loss function. SmoothL1Loss is advantageous in that it can be used to approximate the L1 Loss (mean absolute error) when the error is small and the L2 Loss (mean squared error) when the error is large. This property can reduce the influence of outliers while ensuring the convergence speed, thereby improving the stability and robustness of the model.

Subsequent to the training of the surrogate model, the searching process of the antenna physical parameters is initiated. During this phase, the algorithms autonomously identify better physical parameters by invoking the trained surrogate model and defining the objective function. This approach eliminates the necessity for manual process, which can be efficient and cost-effective. The comprehensive experimental workflow is depicted in Fig. 4.

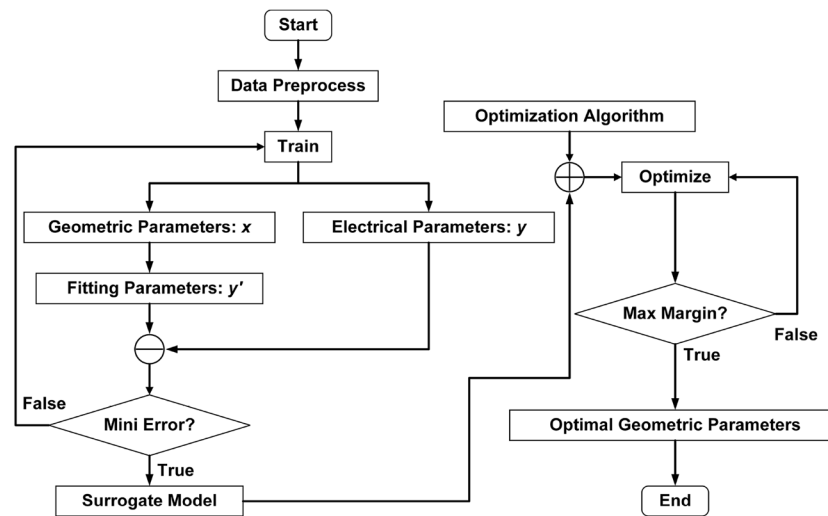


Fig. 4 Experiment process of training and optimization

In the surrogate-assisted pipeline, the optimization operates in the normalized design space (Min-Max scaling consistent with surrogate training). A specific description of the loop is as follows:

- (1) initializing a population of candidate normalized physical vectors;
- (2) evaluating the entire population in-batch via the surrogate;
- (3) computing each candidate's scalar fitness according to the engineering objective, including the margin and penalty;
- (4) updating the population according to the chosen optimizer's rules and enforcing bounds/constraints;
- (5) recording best-so-far and termination criteria;
- (6) repeating the process possibly with multiple independent restarts.

After searching for better physical parameters, the candidates are deduplicated and validated by high-fidelity HFSS simulations, which can save much time and computational cost than the conventional method using full-frequency simulation of HFSS; HFSS results are used as the ultimate decision criterion.

The experimental setup involved a computing device equipped with an Intel i7-13700 H (@2.4 GHz) CPU, an Nvidia RTX 4060 GPU, and 32 GB of memory. The surrogate models are developed via the PyTorch 2.1.0 framework, and the programming language used is Python 3.9.

4.3 Results analyses of the DL surrogate model

To verify the performance of the two models in the S_{11} curve prediction task, Deep Multi-Layer Perceptron (DMLP) [30] and One-Dimensional Convolutional Neural Network (1D-MCNN) [31] from our team lab are selected as baseline models to conduct comparative analyses. The experiment sets the maximum training epoch for each DL model to 1000 epochs, and records the changing trends of training loss and test loss during the training process, as shown in Fig. 5.

A comparison of the loss curves reveals that in the initial training stage, i.e., when the epoch is less than 50, the training loss and test loss values of 1D GP-HetCNN and 1D

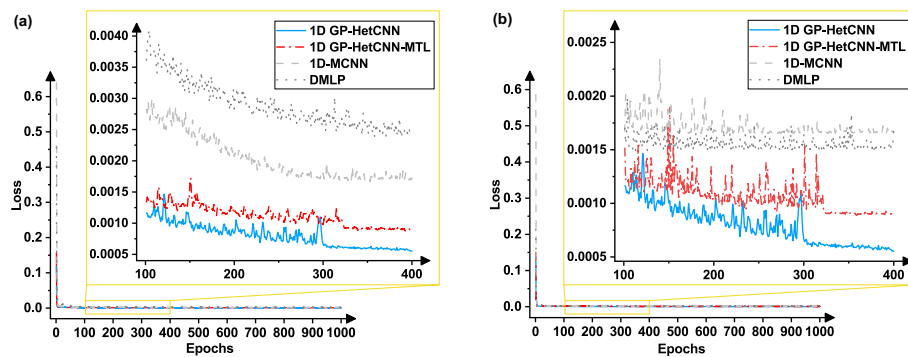


Fig. 5 Training losses and test losses of four models: (a) training losses of 1D GP-HetCNN, 1D GP-HetCNN-MTL, 1D-MCNN and DMLP; (b) test losses of 1D GP-HetCNN, 1D GP-HetCNN-MTL, 1D-MCNN and DMLP

Table 3 Performance comparison of the three models

Surrogate models	Training Loss	Test Loss	RMSE	MAE
1D GP-HetCNN	5.23e−4	4.81e−4	0.0305	0.0189
1D GP-HetCNN-MTL	9.09e−4	9.10e−4	0.0426	0.0282
1D-MCNN	1.71e−3	1.68e−3	0.323	0.267
DMLP	1.96e−3	1.51e−3	0.395	0.286

Note: the RMSE and MAE of 1D-MCNN and DMLP are derived from the papers that have been published publicly. The RMSE and MAE of 1D-MCNN [31]. The RMSE and MAE of DMLP [30]

GP-HetCNN-MTL are relatively low, remaining below 0.2. This observation indicates that the proposed models exhibit a satisfactory initial fitting ability in the early training stage, and the learning process is more stable.

In the subsequent phase of training, as illustrated in the magnified segment of Fig. 5, the training losses and test losses of the two proposed models exhibit a more pronounced downward trend, with a more notable reduction in loss than the DMLP and the 1D-MCNN do.

In the final convergence stage, the loss values of 1D GP-HetCNN and 1D GP-HetCNN-MTL converge to a lower level, far lower than those of DMLP and 1D-MCNN, indicating that the two models have higher learning accuracy in the S_{11} curve prediction task.

The experimental results provide a quantitative assessment of the training effects of different models. As presented in Table 3, after the training phase, the training loss and test loss of the 1D GP-HetCNN are both averagely lower than $6e-4$; the training loss and test loss of the 1D GP-HetCNN-MTL are both averagely lower than $1e-3$. In contrast, the loss values of DMLP (training loss = $1.71e-3$; test loss = $1.68e-3$) and 1D-MCNN (training loss = $1.96e-3$; test loss = $1.51e-3$) are significantly higher.

Moreover, employing the root mean square error (RMSE) and mean absolute error (MAE) as metrics for evaluating model prediction performance reveals that the RMSE and MAE of the two models are all averagely below 0.05, which is significantly lower than those of the DMLP (RMSE = 0.323; MAE = 0.267) and the 1D-MCNN (RMSE = 0.395; MAE = 0.286). This further substantiates the superior accuracy of the proposed models in the S_{11} prediction task.

Furthermore, because the losses of 1D GP-HetCNN-MTL during the training process are combined with three parallel tasks, the experimental results provided in Table 3 show that 1D GP-HetCNN-MTL attains a little higher training/test loss and smaller RMSE/MAE on the integrated frequency band of the S_{11} curve than the MTL version.

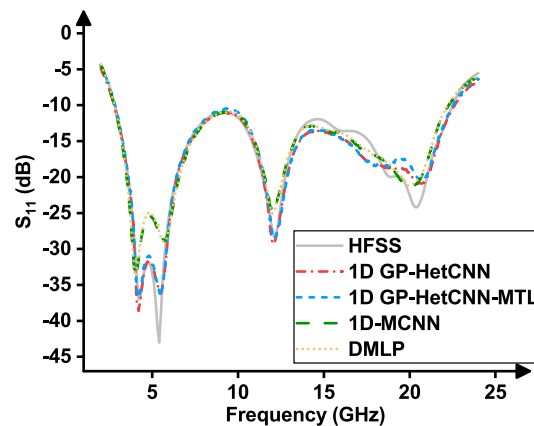


Fig. 6 Original and predicted S_{11} curves from HFSS and four surrogate models (original curve from HFSS and four predicted curves respectively from 1D GP-HetCNN, 1D GP-HetCNN-MTL, 1D-MCNN and DMLP)

To assess the model's capacity to predict samples outside the dataset, as illustrated in Fig. 6, the experiment uses three models to predict the S_{11} curve of a set of antenna physical parameters not included in the dataset. These predicted curves are then compared with the HFSS simulation curves.

As illustrated in Fig. 6, the S_{11} curves predicted by the two proposed models demonstrate a better degree of fitting, both aligning closely with the simulation curve of HFSS. In the low-frequency range close to 5 GHz, four models all predict two minimum points of the curve, but the predicted curves of 1D GP-HetCNN and 1D GP-HetCNN-MTL are much closer to the original curve of HFSS than 1D-MCNN and DMLP.

In the middle-frequency range close to 12 GHz, 1D GP-HetCNN and 1D GP-HetCNN-MTL show better predicting performance, as both predicted curves obtain lower S_{11} minimum values which are almost equal to that of HFSS.

Yet, due to the unstable changing trend of S_{11} values in the high-frequency range from 17GHz to 23GHz of the dataset, it is a great challenge for the four models to capture highly precise features of this range; thus, some deviations are provided between the predicted curves and the original curve. But from Fig. 6, there are still more complicated changing features of more than one minimum point of the curve that are predicted by the two proposed models, thus remaining competitive with other models.

4.4 Results analyses of the optimization methods

In the optimization stage, the performance of WOA and COA in antenna parameter optimization tasks is provided. The Bayesian optimization (BO) algorithm [32], the particle swarm optimization (PSO) algorithm [33–35], and the gray wolf optimization (GWO) algorithm [36, 37] are used as baseline algorithms. All algorithms search for better antenna physical parameters by calling the two trained surrogate models proposed in this paper. Every algorithm runs 30 times independently. After running and deduplication process, only 44 groups of geometric parameters need to be validated using HFSS during our experiment.

To ensure a fair and comprehensive evaluation of the performance of each algorithm, setting the number of predictions of the surrogate model as the total function-evaluation budget which is equalized (2000) across population-based methods (WOA, COA, GWO and PSO) and BO. The experiment analyzes three key indicators: average optimization

Table 4 Performance comparison of each method

Methods	Time (s)	Bandwith (GHz)	FBW
HFSS	-	19.3	155.02%
1D GP-HetCNN-WOA	2.82	19.5	156.03%
1D GP-HetCNN-COA	2.77	19.5	155.77%
1D GP-HetCNN-GWO	2.88	19.6	155.95%
1D GP-HetCNN-PSO	2.89	19.6	156.01%
1D GP-HetCNN-BO	7653.75	19.6	156.61%
1D GP-HetCNN-MTL-WOA	3.28	20.0	156.22%
1D GP-HetCNN-MTL-COA	3.31	20.0	156.16%
1D GP-HetCNN-MTL-GWO	3.30	20.0	156.33%
1D GP-HetCNN-MTL-PSO	3.32	20.0	156.52%
1D GP-HetCNN-MTL-BO	7020.34	19.9	156.56%

time, average absolute operating bandwidth verified from HFSS, and average fractional bandwidth (FBW), as shown in Table 4.

In comparison with HFSS (bandwidth = 19.3 GHz; FBW = 155.02%), each method obtain a wider bandwidth and a higher FBW, indicating the effectiveness of our methods of combining the surrogate models and intelligent optimization methods.

First of all, the time costs of BO combined with two surrogate models are obviously higher than those of other methods, even if the bandwidths (1D GP-HetCNN-BO: 19.6 GHz; 1D GP-HetCNN-MTL-BO: 19.9 GHz) and FBWs (1D GP-HetCNN-BO: 156.61%; 1D GP-HetCNN-MTL-BO: 156.56%) are all competitive.

Secondly, except for the two methods of BO combined with two surrogate models, the time costs of other algorithms combined with 1D GP-HetCNN-MTL are a little higher than those of other algorithms combined with 1D GP-HetCNN, but other algorithms combined with 1D GP-HetCNN-MTL can generally obtain higher bandwidths and FBWs, especially the methods of WOA and COA combined with 1D GP-HetCNN-MTL show the widest bandwidth (20.0 GHz) of all methods.

Moreover, each indicator of the methods of GWO and PSO combined with two surrogate models are competitive, too. The time cost, bandwidth and FBW are all close to those of the methods of WOA/COA combined with two surrogate models.

Figure 7 provides the boxplot and the empirical cumulative distribution function (ECDF) of different methods. The boxes indicate interquartile range (IQR), central lines denote medians, whiskers extend to the most extreme non-outlier values (Tukey rule, $1.5 \times \text{IQR}$), and individual markers denote mean values (rectangular points) and outliers (rhombic points).

The detailed distributional statistics, including the mean $\pm$ standard deviation (SD), median, IQR and outliers of all methods, are shown in Table 5.

For one thing, comparing five methods combined with 1D GP-HetCNN:

-WOA attains the largest central values (mean $\pm$ SD: 15.296 ± 0.0482 ; median: 15.275), indicating that across runs it tends to find the best solutions on average. GWO and COA achieve intermediate central performance (means and medians slightly below WOA), while BO is close to those but slightly lower. PSO yields substantially lower central values (mean $\pm$ SD: 15.087 ± 0.0327), indicating poorer final performance in this experiment.

-GWO shows the smallest interquartile ranges (IQRs: 0.0200), implying that when it converges, GWO is more consistent (narrow box). COA also shows a relatively small spread (IQR: 0.0300) and around a higher median, meaning that COA is stable and

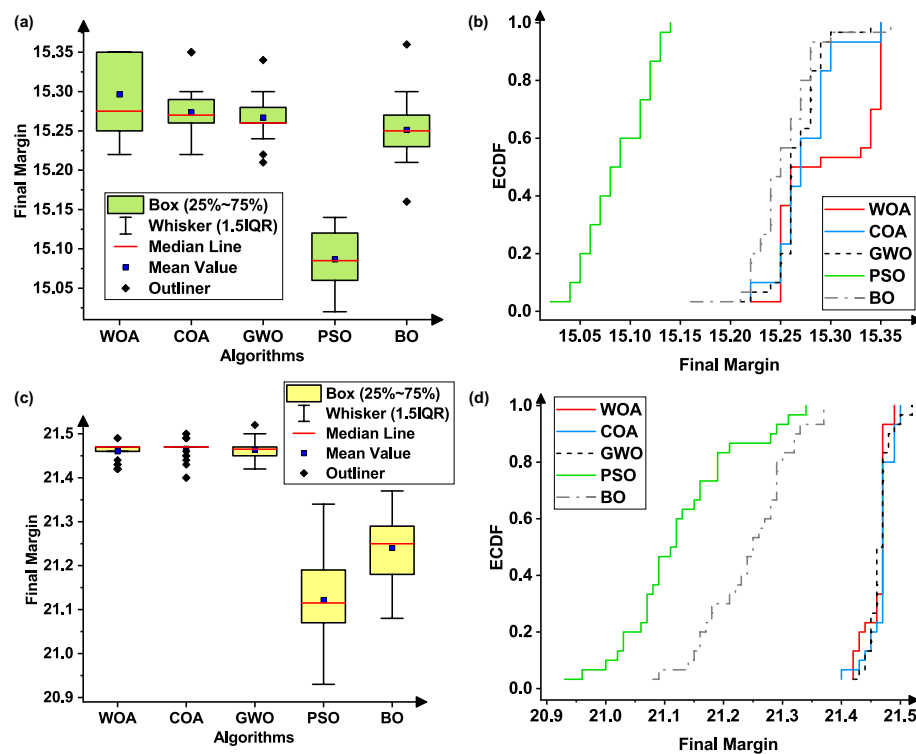


Fig. 7 Bosplots and ECDFs of different methods: (a) boxplot of five algorithms combined with 1D GP-HetCNN; (b) ECDF of five algorithms combined with 1D GP-HetCNN; (c) boxplot of five algorithms combined with 1D GP-HetCNN-MTL; (d) ECDF of five algorithms combined with 1D GP-HetCNN-MTL

Table 5 Detailed distributional statistics of each method

Methods	Mean $\pm$ SD	Median	IQR	Outlier
1D GP-HetCNN-WOA	15.296 $\pm$ 0.0482	15.275	0.100	-
1D GP-HetCNN-COA	15.273 $\pm$ 0.0309	15.270	0.0300	15.35
1D GP-HetCNN-GWO	15.267 $\pm$ 0.0239	15.260	0.0200	3 outliers[1]
1D GP-HetCNN-PSO	15.087 $\pm$ 0.0327	15.085	0.0575	-
1D GP-HetCNN-BO	15.251 $\pm$ 0.0346	15.250	0.0375	2 outliers[2]
1D GP-HetCNN-MTL-WOA	21.460 $\pm$ 0.0205	21.470	0.0100	4 outliers[3]
1D GP-HetCNN-MTL-COA	21.466 $\pm$ 0.0236	21.470	0.000100	7 outliers[4]
1D GP-HetCNN-MTL-GWO	21.464 $\pm$ 0.0196	21.465	0.0175	21.52
1D GP-HetCNN-MTL-PSO	21.122 $\pm$ 0.0989	21.115	0.113	-
1D GP-HetCNN-MTL-BO	21.240 $\pm$ 0.0755	21.250	0.110	-

Note: Specific values of outliers. 1D GP-HetCNN-GWO: three outliers which are 15.21, 15.22 and 15.34 respectively. 1D GP-HetCNN-BO: two outliers which are 15.16 and 15.36 respectively. 1D GP-HetCNN-MTL-WOA: four outliers which are 21.42, 21.43, 21.44 and 21.49 respectively. 1D GP-HetCNN-MTL-COA: seven outliers which are 21.40, 21.43, 21.44, 21.45, 21.46, 21.49 and 21.50 respectively

consistently produces better solutions. WOA exhibits a larger IQR (0.100), indicating greater variability, which means that it sometimes finds very good solutions but has more run-to-run spread.

-BO shows two Tukey outliers (15.16, 15.36), which suggests it occasionally finds poorer/better-than-typical runs (long right tail). GWO has two low-side outliers (15.21, 15.22), indicating an occasional poor run. Also, COA shows one high-side outlier (15.35); while WOA shows no Tukey outliers by the $1.5 \times \text{IQR}$ rule in these samples.

For another thing, comparing five methods combined with 1D GP-HetCNN-MTL:

-COA attains the largest central values (mean $\pm$ SD: 21.466 ± 0.0236 ; median: 21.470), indicating that across runs it tends to find the best solutions on average. WOA and GWO achieve slightly lower but competitive central performance (WOA: 21.460 ± 0.0205 ; WOA: 21.464 ± 0.0196), while BO and PSO yield substantially lower central values (BO: 21.240 ± 0.0755 ; PSO: 21.122 ± 0.0989), indicating weaker final performance under the present experimental settings.

-In this experiment, COA exhibits the smallest interquartile range (IQR: 0.000100), implying super tight run-to-run consistency around a high median; WOA (IQR: 0.0100) and GWO (IQR: 0.0175) show modest spread, while BO and PSO show substantially larger spread (IQRs: 0.110 and 0.113 respectively), meaning these algorithms are more variable in final performance and typically produce lower-quality solutions. In short, COA is both the best in central tendency and the most concentrated; BO/PSO are lower and more dispersed.

-However, COA shows the most outliers, indicating occasional extreme behaviour in both directions. WOA also shows four outliers; while no clear outliers are present for PSO and BO by the $1.5 \times \text{IQR}$ rule in these samples. These observations suggest that while COA is typically the strongest and most consistent, it can occasionally produce extreme results (both unusually good and poor); BO and PSO are more stably mediocre under current settings.

The present study is intentionally scoped to develop and validate lightweight, parameter-efficient surrogates and to demonstrate an end-to-end surrogate-assisted optimization pipeline. Also, because of the high cost of conventional HFSS full-frequency simulation, the size of the dataset in the study is light, thus it may require additional validation and likely modest methodological extensions before being generalized to noisy, real-world measurements.

5 Conclusion

In this study, a lightweight surrogate model—1D GP-HetCNN, and its MTL version—1D GP-HetCNN-MTL, are proposed for predicting the S_{11} curve of the antenna. The surrogate models can quickly and accurately predict the curve by inputting physical parameters of the antenna. The integration of antenna modeling and DL is a significant contribution of this paper. Subsequently, two antenna optimization methods based on the WOA and COA are proposed respectively. On the basis of two surrogate models, the two methods automatically and efficiently search for physical parameters that broaden the operating bandwidth of the antenna, achieving the goal of intelligent and automatic antenna optimization. Moreover, more studies will be set in the future, including EM multi-objective design and optimization of antennas, the improvement and combination of various intelligent optimization algorithms, the joint training of the surrogate models and Intelligent optimization methods, and mitigating the potential interference factors arising during the fabricating process when using neural networks for antenna design.

Author Contributions

Ximin Yang made substantial contributions to the design of the work and drafted the work. Jingchang Nan revised the work critically for important intellectual content. Minghuan Wang provided the simulation file for creating the dataset.

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Data Availability

The dataset used and analysed during the current study available from the corresponding author on reasonable request.

Declarations

Conflict of interest

The authors declare no conflict of interest.

Ethical Approval

Not applicable.

Consent to participate

Not applicable.

Consent to publish

Not applicable.

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